

Tab 1



INTERNSHIP & JOB PLACEMENT HUB

FUNCTIONAL SPECIFICATIONS

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Authorization Memorandum

I have carefully assessed the Functional Specifications Document for the Internship & Job Placement Hub.

MANAGEMENT CERTIFICATION - Please check the appropriate statement.

_____ The document is accepted.

_____ The document is accepted pending the changes noted.

_____ The document is not accepted.

We fully accept the changes as needed improvements and authorize initiation of work to proceed. Based on our authority and judgment, the continued operation of this system is authorized.

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Add other names and roles as necessary.

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1 Preface

The Job and Internship System has been developed to address the challenges faced by students and graduates in accessing career opportunities, as well as the difficulties employers experience in finding suitable candidates efficiently. Many graduates struggle to discover and apply for relevant jobs or internships, while employers often face lengthy and inefficient recruitment processes.

This platform provides a centralized, structured environment where students and graduates can create profiles, explore and apply for opportunities, and track their applications. Employers can post job and internship listings, review candidate applications, and gain insights into hiring trends. By integrating these features into one system, it enhances employability, supports workforce development, and strengthens the connection between education and employment.

1.1 Purpose

The purpose of this Functional Specifications Document is to define the expected functionalities and features of the Job and Internship System. While the Technical Specifications Document focuses on how the system will be built, this document focuses on what the system will do from a functional perspective.

It ensures that all stakeholders including developers, project managers, and end-users have a clear understanding of the system's intended behavior, capabilities, and boundaries. The document outlines the user-facing features, workflows, data interactions, and performance expectations, serving as a reference point throughout the design, development, and testing phases.

By establishing a shared understanding of functionality, the document helps align stakeholder expectations on what the system will achieve and how users will interact with it, guide development by clearly defining workflows, inputs, outputs, and user interactions, ensure comprehensive coverage of all functional requirements, and provide a framework for testing and validation to confirm that the system meets its intended purpose.

1.2 Intended Audience and Pertinent Sections

This document is intended for several groups of stakeholders, each with areas most relevant to them:

- Project Sponsors will be most interested in the Preface and Introduction, which outline the project's vision and goals.

- Project Managers will rely on the Preface, Introduction, and Requirements sections to oversee planning and implementation.
- Developers will focus on the Requirements, System Features, and Technical Environment to guide the building of the system.
- End Users such as students, graduates, and employers, will primarily engage with the features, though they may also benefit from the overview provided here.

1.3 Project Scope

The Job and Internship System aims to deliver a centralized platform where students and graduates can discover opportunities, while employers can find suitable candidates. The planned scope includes:

1. User registration and secure login
2. Profile creation and management (for candidates and employers)
3. Posting and browsing of jobs and internships
4. Application submission and tracking
5. Employer review and feedback on candidates
6. Search and filtering functions to improve job matching
7. Notifications and alerts for new opportunities and application updates
8. Basic analytics to track applications and postings

1.4 Reference Documents

Internship&JobPlacementHubBC&FS.docx

Internship&JobPlacementHubTechnicalSpecifications.docx

Internship&JobPlacementHubFunctionalSpecifications.docx

1.5 Abbreviations and Acronyms

POPIA – Protection of Personal Information Act

CV – Curriculum Vitae

HR – Human Resources

AWS – Amazon Web Services

GCP – Google Cloud Platform

SMS – Short Messaging System
API – Application Programming Interface
REST API – Representational State Transfer API

1.6 Document Conventions

This document follows these conventions:

- Heading levels are indicated by heading styles (Heading 1, Heading 2, etc.)
- Bulleted lists are used to present sets of related items or options.
- Numbered lists are used to present sequential steps or instructions.
- Code snippets or file names are formatted in a monospace font.
- Important notes or tips are highlighted using a blockquote style.

2 GENERAL DESCRIPTION

2.1 Product Context

In South Africa, a lot of graduates leave university with qualifications but then struggle to get into the job market. One of the big issues is that their CVs are often not standardised, sometimes missing key details, or even hard for employers to trust. On the other side, companies spend time going through piles of CVs that don't always give them what they need.

The idea behind the Graduate Placement Portal is to fix this gap by giving graduates one place where they can create proper profiles, upload and verify their documents, and apply for jobs directly. Employers also benefit because they can quickly find reliable information instead of digging through unreliable CVs.

Put simply, the portal is meant to make life easier: graduates can focus on presenting themselves well, and employers can make faster and better hiring decisions.

2.2 User Classes and Characteristics

Primary Users – Students and Graduates

- Mostly final-year students or recent graduates.
- Different levels of computer skills (some advanced, some basic).
- Eager to start their careers and need a way to show their qualifications clearly.

What they need:

- A simple platform to create/update a profile.
- A safe way to store and verify documents.
- The option to apply for jobs directly without extra steps.

Secondary Users – Employers / Recruiters

- HR staff, hiring managers, or recruitment agents.
- Don't have time to sort through unverified or inconsistent CVs.
- Need a reliable way to filter and find the right candidates.

What they need:

- Quick candidate search and filtering tools.
- Easy access to verified qualifications.
- A smoother way to track applications.

Future Users – Institutions and Career Services

- Universities and colleges wanting to see how their graduates perform in the job market.
- Career centres that support students with placements.
- Alumni groups tracking long-term career outcomes.

2.3 Overview of Functional Requirements

Create Job/Internship Listing:

- Employers can create a job/internship post.
- Functionality accepts job title, description, required skills, location, employment type (full-time, part-time, internship) and application deadline.
- Applications are stored in the database.
- Confirmation message is displayed, and new listing is visible to relevant prospective applicants.

Edit Job/Internship Listing:

- Employers can modify existing job or internship posts.
- Functionality allows updates to job title, description, required skills, location, employment type, or deadline.
- Changes and updates should be validated before reflecting.
- Edited listings reflect changes in real-time on the web application.

Delete Job/Internship Listing:

- Employers can remove job or internship posts.
- Functionality prompts for confirmation before a post is deleted.
- The system removes the listing from the database and updates the student view to reflect the removal.

Search Job/Internship Listings:

- Students can search for job or internship opportunities based on their skills or field of study.
- Functionality allows searching by job title, company, location, skills or employment type.
- The system searches for matching results in the database and displays matches.

Submit Application

- Students can apply for jobs or internships via the system.
- Functionality allows uploading CV's and additional documents.
- The system validates uploaded documents and notify the relevant employer of new applications.
- Students receive a confirmation message when application is successful, or error if the application is unsuccessful.

Track Application Status

- Students can monitor the application status on the system.
- Functionality provides updates on application (received, accepted, rejected).
- The application status is displayed on the system.

User Registration:

- Students and employers can create accounts to access the application.
- Functionality requires input of user type (student or employer), email address, password and relevant details (for example the student's qualifications or the employer company name).
- The system validates input and creates the account in the database.

User login:

- Users can access their accounts.
- Functionality accepts the user email and password as prompts.
- The system validates user credentials against the database before system access is granted.

User Profile Management:

- Users can view and edit their profile.

- Functionality allows students to update skills, field of study, location, or Curriculum Vitae.
- The system validates changes and updates the user profile in the database.

Employer Verification:

- Administrators can verify employer accounts for safety and security purposes.
- Functionality allows employers to submit verification documents (for example B-BBEE certificates, company registration).
- The system stores documents, and administrators approve or reject verification documents.
- Verified employers are given access to post listings and communicate with applicants.

2.4 Overview of Data Requirement

- Student data: Personal information, academic background, supporting documents like CV, skills
- Transactional data: Applications, system notifications for students, employer feedback, interview outcomes
- Employer data: company information, available jobs, job requirements, data to verify whether the company/employer is legit
- System metadata: login history, placement data, engagement metrics

2.5 Operating Environment

- **Frontend:** Web browsers
- **Backend:** Cloud-hosted (AWS/Azure/GCP), scalable architecture.
- **Database:** PostgreSQL or Firebase Firestore with secure storage.

2.6 General Constraints, Assumptions, Dependencies, Guidelines

Constraints:

- **Project Scope:** The development will prioritize core functionalities as defined in the project scope document. Additional features may be considered in future iterations based on feasibility and user feedback.
- **Budget:** Development will adhere to the allocated budget. This might influence the complexity of certain functionalities or require creative solutions within the financial constraints.
- **Timeline:** The portal will be developed within the established timeframe as outlined in the project schedule

Assumptions:

- **User Technical Literacy:** Users will possess basic smartphone literacy and be comfortable navigating around the portal.
- **Device Compatibility:** The ideal usage of the portal will be on a computer or laptop. A compatibility list might be created based on research and testing.
- **Data Security:** Users understand the importance of protecting their login credentials and collection data. Security measures will be implemented, but user awareness plays a role as well.

Dependencies:

- **Development Tools and Libraries:** The portal development will rely on specific software development tools and libraries compatible with the Windows platform. These will be chosen based on their suitability for the functionalities and development efficiency.
- **User Data Storage:** The portal will depend on a secure data storage solution to manage user accounts and collection information.

Guidelines:

- **User Interface Design:** The portal will follow established Windows design guidelines to ensure a familiar and intuitive user experience for computer and or laptop users.
- **Accessibility:** The portal will strive to be accessible to users with disabilities, following relevant accessibility standards for mobile applications.
- **Testing and Quality Assurance:** The development process will incorporate thorough testing procedures to ensure the portal functions as intended and delivers a high-quality user experience.

2.7 Design and Implementation Constraints

The design and implementation of the system will adhere to the following constraints:

- Regulatory and Compliance constraints:

The system must comply with national and international laws governing data collection, storage, sharing, and security. Since the portal will serve South African universities and students, compliance with POPIA is mandatory.

The system also needs an employer verification workflow.

- Technical constraints:

The technical constraints ensure the system is secure, scalable, fast, and compliant while still being usable on common devices. They restrict the tech stack choices but also guarantee reliability, performance, and institutional adoption.

The system must be developed as a secure, scalable, and cloud-native web application. It requires a responsive design that complies with accessibility standards and supports multiple languages. The architecture must be built on a relational database for structured data integrity and be capable of handling high concurrency with low response times. Mandatory security measures include end-to-end encryption, robust role-based access control, and integration with institutional single sign-on.

- Design constraints:

The design of the Internship and Job Placement Portal must prioritize responsiveness, accessibility, and simplicity to ensure usability across different devices. It must also remain lightweight and data-efficient to accommodate users with limited connectivity. Security and trust are one of the important parts of the system, the system design must have security features indicating that it is secure.

- Security constraints

The security constraints ensure that the portal protects student trust, employer credibility, and institutional reputation. By enforcing strict authentication, encryption, fraud prevention, auditing, and compliance, the system minimizes risks of data breaches, scams, and unauthorized access.

- Budget constraints

The design and implementation should consider the allocated budget for the system. It should be cost – efficient and scalable. The system should implement features like user-friendly interfaces, affordable cloud hosting and low maintenance design to reduce costs and stay within the budget.

2.8 User Documentation

An online user guide will be provided for students, employers and administrators. They will also be provided with help tools and functions within the application to provide easy access and navigation.

3 REQUIREMENTS

3.1 External Interface Requirements

- Integration with **email/SMS APIs** for notifications.
- Integration with **cloud storage APIs** for CV/document handling.
- Secure REST APIs for authentication and data exchange.

3.2 User Interfaces

Core UI components:

- Login & Registration Screens
- Student Dashboard (CV upload, job applications)
- Employer Dashboard (job posting, applicant review)
- Admin Dashboard (verification, analytics)
- Search & Filter Screen (jobs/internships)
- Messaging Interface

3.3 Hardware Interfaces

- Compatible with laptops, tablets, and smartphones.
- Uses camera/microphone only for future features (video interviews).

3.4 Software Interfaces

Real time Job Dashboard that is filtered based on user selection that also displays details of each job such as placements, active users, job specifications. These should also be unlisted once all positions are filled, only displaying to users that have been accepted, maintaining open communication between employer and employee. This should be user friendly as well as accommodating for users looking for work as

well as employers, to ensure ease of use and access.

3.5 Communications Interfaces

Users will need to login and register, as well as see a dynamic job listings board and messaging feature to allow them to communicate with employers of listings that the users are interested in.

3.6 Functional Requirements

1. Job and Internship Posting

Purpose/Description:

Enable employers to create and manage job or internship postings by filling out details such as title, description, requirements, and deadlines.

Inputs:

- Employer login credentials
- Job posting details (title, description, requirements, deadline, etc.)

Processing:

- Authenticate employer account
- Save posting details in the database
- Validate input fields (mandatory fields, deadlines not in the past, etc.)

Outputs:

- Job/internship posting visible to applicants
- Confirmation message for successful posting

2. Viewing Opportunities

Purpose/Description:

Allow applicants to browse available opportunities.

Inputs:

- Applicant login credentials
- Search/filter terms for opportunities

Processing:

- Authenticate applicant account

- Filter job postings based on search criteria

Outputs:

- Display of available postings

3. Applying for Positions

Purpose/Description:

Allow students/applicants to apply for positions by submitting required documents and linking applications to job postings.

Inputs:

- Applicant login credentials
- CV/resume
- Cover letter (and other supporting documents if required)

Processing:

- Authenticate applicant account
- Validate submitted documents (file type, size, completeness)
- Link application to the specific job posting
- Notify employer of new application

Outputs:

- Confirmation message for successful submission
- Application visible in employer dashboard

4. CV/Profile Management

Purpose/Description:

Provide applicants with the ability to upload, edit, and manage CVs, profiles, and relevant documents.

Inputs:

- Applicant login credentials
- File uploads (CVs, certificates, supporting docs)
- Profile details (skills, education, experience, etc.)

Processing:

- Authenticate applicants
- Validate and store files in secure storage
- Update applicant profile data in the database

Outputs:

- Updated applicant profile viewable in dashboard
- Confirmation of successful upload or edit

5. Posting Approval/Rejection**Purpose/Description:**

Allow administrators to review and approve or reject employer job postings.

Inputs:

- Admin login credentials
- Employer posting details

Processing:

- Authenticate admin account
- Display pending postings
- Update posting status to approved/rejected
- Record action in activity logs

Outputs:

- Posting made visible to applicants (if approved)
- Notification to employer about status

6. Applicant Management (Employers)**Purpose/Description:**

Enable employers to view and filter applicants per posting.

Inputs:

- Employer login credentials
- Search/filter criteria (skills, location, education, etc.)

Processing:

- Authenticate employer
- Retrieve list of applicants linked to posting
- Apply filtering based on selected criteria

Outputs:

- Filtered list of applicants
- Option to export or view applicant details

7. Secure Authentication

Purpose/Description:

Ensure role-based secure login for Admins, Employers, and Applicants.

Inputs:

- User login credentials (username/email + password)
- Password recovery details (e.g., email or phone number)

Processing:

- Encrypt and validate login credentials
- Verify user role (Admin/Employer/Applicant)
- Provide password reset options if needed

Outputs:

- Successful login and access to appropriate dashboard
- Error message for invalid credentials

8. Search and Filter for Opportunities

Purpose/Description:

Provide advanced search and filtering for job/internship opportunities.

Inputs:

- Search terms (keywords, job title)
- Filters (location, deadline, requirements, etc.)

Processing:

- Query job database with given filters
- Return only matching records

Outputs:

- Search results displayed to user
- Option to refine or reset filters

9. Accessibility and Responsive Design

Purpose/Description:

Ensure the system is usable across devices and accessible for users with disabilities.

Inputs:

- User device/browser settings
- Accessibility preferences

Processing:

- Apply responsive design for different screen sizes
- Implement accessibility features (labels, alt text, keyboard navigation)

Outputs:

- Optimized user interface on all devices
- Compliance with accessibility standards

10. Scalability and Performance

Purpose/Description:

Ensure system scales with growing users and data load.

Inputs:

- Increasing number of user requests
- Expanding job postings and applicant data

Processing:

- Store data in scalable database/cloud hosting
- Perform load balancing and optimization
- Conduct performance tests under load

Outputs:

- Stable and responsive system under high traffic
- System health and performance reports

3.7 Performance Requirements

- **Responsiveness:**
The app should respond quickly when users navigate between pages, search for jobs, or upload documents.
- **Efficiency:**
It should run smoothly on normal laptops, desktops, and mobile phones without needing very high system specifications.
- **Offline Functionality:**
Users should be able to view already saved information (like their profile or previously seen job ads) even if the internet is down, with changes syncing later.

- **System Performance:**
The platform should allow multiple students and employers to use the system at the same time without slowing down noticeably.
- **Error Handling:**
If something goes wrong (e.g., failed login, file upload error), the system should show clear messages and guide the user on what to do next.
- **Scalability (Basic):**
The system design should allow adding more features or supporting more users later without needing a full rebuild.

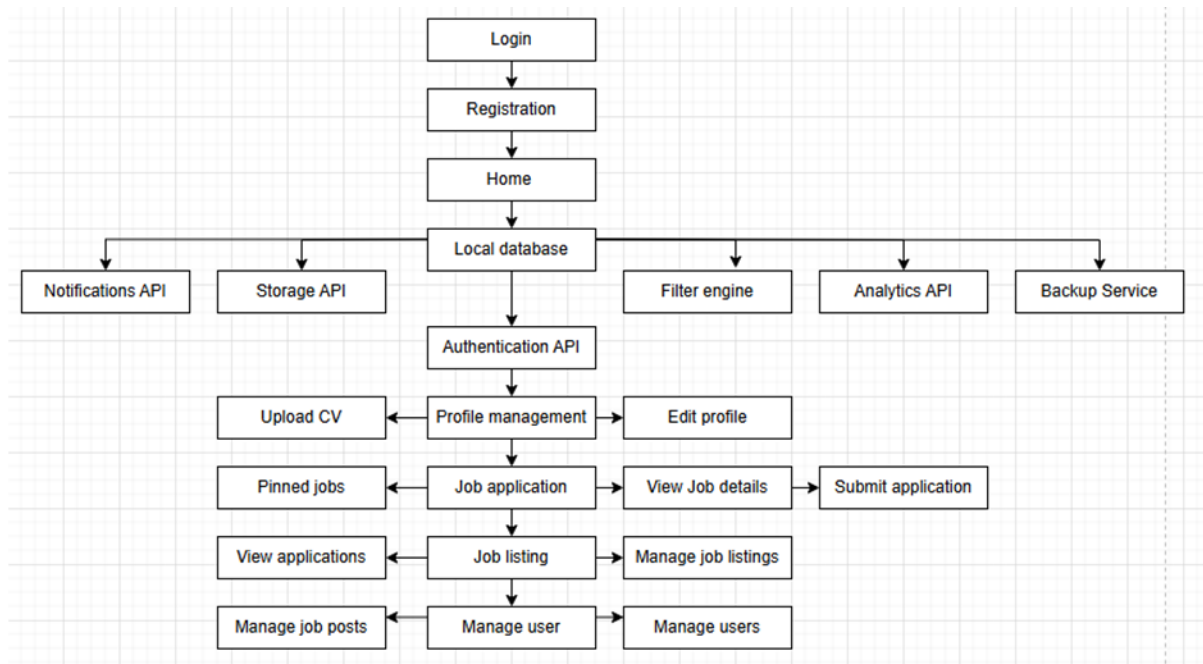
3.8 Security

- **Data Security:**
User details such as login information, CVs, and job applications should be stored in a way that prevents unauthorized access such as encrypted storage.
- **API Communication Security:**
Any connection with other services (e.g., emails or notifications) should not expose sensitive information.
- **User Authentication:**
A login system must separate students, employers, and administrators so that each can only access their own features.
- **Password Protection:**
Passwords must not be visible to anyone, including system administrators. A simple hashing technique can be used to protect them.
- **Session Management:**
Users should be logged out after a certain period of inactivity to prevent unauthorized use.
- **Backup of Data:**
Regular database backups should be done so that no important user data is lost if the system fails.

3.9 Usability

- **Intuitive Interface:**
The app should be simple and easy to use for people with different levels of computer knowledge.
- **Task Efficiency:**
Posting a job, applying for a position, or editing a profile should take only a few simple steps.
- **Discoverability:**
Important features such as “Apply,” “Upload CV,” or “Post Job” should be clearly visible and easy to find.
- **Accessibility:**
The system should work properly on both desktop and mobile devices.
- **Consistency:**
The same design style (colors, buttons, navigation) should be used across all pages so users do not get confused.
- **Feedback to Users:**
Whenever a task is completed (e.g., application submitted, job posted), the system should confirm it with a success message.
- **Help and Guidance:**
A small help section, tooltips, or instructions should be available to assist new users in understanding how to use the platform.

3.10 Data Flow



3.11 Other Requirements

There are no other requirements.

